```
================================================================================
Q1) What are the five most similar segments to segment '267:476'?

Query:
select p.title, s.id, s.content, s.start_time, s.end_time,
  s.embedding <-> q.embedding as distance
  from segment s
  join podcast p on s.podcast_id = p.id
  join segment q on q.id = '267:476'
  where s.id <> q.id
  order by distance
  limit 5;

-- result 1 --
     title: Ryan Graves: UFOs, Fighter Jets, and Aliens | Lex Fridman Podcast #308
        id: 113:2792
   content:  encounters, human beings, if we were to meet another alien
start_time: 6725.6200
  end_time: 6729.8600
  distance: 0.6483
-- result 2 --
     title: Richard Dawkins: Evolution, Intelligence, Simulation, and Memes | Lex
Fridman Podcast #87
        id: 268:1019
   content:  Suppose we did meet an alien from outer space
start_time: 2900.0400
  end_time: 2903.0800
  distance: 0.6558
-- result 3 --
     title: Jeffrey Shainline: Neuromorphic Computing and Optoelectronic
Intelligence | Lex Fridman Podcast #225
        id: 305:3600
   content:  but if we think of alien civilizations out there
start_time: 9479.9600
  end_time: 9484.0400
  distance: 0.6595
-- result 4 --
     title: Michio Kaku: Future of Humans, Aliens, Space Travel & Physics | Lex
Fridman Podcast #45
        id: 18:464
   content:  So I think when we meet alien life from outer space,
start_time: 1316.8600
  end_time: 1319.5800
  distance: 0.6662
-- result 5 --
     title: Alien Debate: Sara Walker and Lee Cronin | Lex Fridman Podcast #279
        id: 71:989
   content:  because if aliens come to us
start_time: 2342.3400
  end_time: 2343.6200
  distance: 0.6743

================================================================================
Q2) What are the five most dissimilar segments to segment '267:476'?

Query:
select p.title, s.id, s.content, s.start_time, s.end_time,
  s.embedding <-> q.embedding as distance
  from segment s
```

```
  join podcast p on s.podcast_id = p.id
  join segment q on q.id = '267:476'
  where s.id <> q.id
  order by distance desc
  limit 5;

-- result 1 --
     title: Jason Calacanis: Startups, Angel Investing, Capitalism, and Friendship
| Lex Fridman Podcast #161
        id: 119:218
   content:  a 73 Mustang Grande in gold?
start_time: 519.9600
  end_time: 523.8000
  distance: 1.6158
-- result 2 --
     title: Rana el Kaliouby: Emotion AI, Social Robots, and Self-Driving Cars |
Lex Fridman Podcast #322
        id: 133:2006
   content:  for 94 car models.
start_time: 5818.6200
  end_time: 5820.8200
  distance: 1.5863
-- result 3 --
     title: Travis Stevens: Judo, Olympics, and Mental Toughness | Lex Fridman
Podcast #223
        id: 283:1488
   content:  when I called down to get the sauna.
start_time: 3709.3400
  end_time: 3711.1000
  distance: 1.5726
-- result 4 --
     title: Jeremy Howard: fast.ai Deep Learning Courses and Research | Lex Fridman
Podcast #35
        id: 241:1436
   content:  which has all the courses pre-installed.
start_time: 4068.9000
  end_time: 4071.1400
  distance: 1.5663
-- result 5 --
     title: Joscha Bach: Nature of Reality, Dreams, and Consciousness | Lex Fridman
Podcast #212
        id: 307:3933
   content:  and very few are first class and some are budget.
start_time: 10648.6400
  end_time: 10650.9600
  distance: 1.5616

================================================================================
Q3) What are the five most similar segments to segment '48:511'?

Query:
select p.title, s.id, s.content, s.start_time, s.end_time,
  s.embedding <-> q.embedding as distance
  from segment s
  join podcast p on s.podcast_id = p.id
  join segment q on q.id = '48:511'
  where s.id <> q.id
  order by distance
  limit 5;
```

```
-- result 1 --
     title: Andrew Huberman: Neuroscience of Optimal Performance | Lex Fridman
Podcast #139
        id: 155:648
   content:  Is there something interesting to you or fundamental to you about the
circuitry of the brain
start_time: 3798.4800
  end_time: 3805.8400
  distance: 0.6523
-- result 2 --
     title: Cal Newport: Deep Work, Focus, Productivity, Email, and Social Media |
Lex Fridman Podcast #166
        id: 61:3707
   content:  of what we might discover about neural networks?
start_time: 8498.0200
  end_time: 8500.1000
  distance: 0.7121
-- result 3 --
     title: Matt Botvinick: Neuroscience, Psychology, and AI at DeepMind | Lex
Fridman Podcast #106
        id: 48:512
   content:  And our brain is there. There's some there's quite a few differences.
Are some of them to you either interesting or perhaps profound in terms of in terms
of the gap we might want to try to close in trying to create a human level
intelligence.
start_time: 1846.8400
  end_time: 1865.8400
  distance: 0.7196
-- result 4 --
     title: Yann LeCun: Dark Matter of Intelligence and Self-Supervised Learning |
Lex Fridman Podcast #258
        id: 276:2642
   content:  Have these, I mean, small pockets of beautiful complexity. Does that,
do cellular automata, do these kinds of emergence and complex systems give you some
intuition or guide your understanding of machine learning systems and neural
networks and so on?
start_time: 8628.1600
  end_time: 8646.1600
  distance: 0.7357
-- result 5 --
     title: Stephen Wolfram: Fundamental Theory of Physics, Life, and the Universe
| Lex Fridman Podcast #124
        id: 2:152
   content:  So is there something like that with physics where so deep learning
neural networks have been around for a long time?
start_time: 610.8600
  end_time: 618.8600
  distance: 0.7367

================================================================================
Q4) What are the five most similar segments to segment '51:56'?

Query:
select p.title, s.id, s.content, s.start_time, s.end_time,
  s.embedding <-> q.embedding as distance
  from segment s
  join podcast p on s.podcast_id = p.id
  join segment q on q.id = '51:56'
```

```
  where s.id <> q.id
  order by distance
  limit 5;

-- result 1 --
     title: George Hotz: Hacking the Simulation & Learning to Drive with Neural
Nets | Lex Fridman Podcast #132
        id: 308:144
   content:  I mean, we don't understand dark energy, right?
start_time: 500.4400
  end_time: 502.6000
  distance: 0.6682
-- result 2 --
     title: Lex Fridman: Ask Me Anything - AMA January 2021 | Lex Fridman Podcast
        id: 243:273
   content:  Like, what's up with this dark matter and dark energy stuff?
start_time: 946.2200
  end_time: 950.1200
  distance: 0.7356
-- result 3 --
     title: Katherine de Kleer: Planets, Moons, Asteroids & Life in Our Solar
System | Lex Fridman Podcast #184
        id: 196:685
   content:  being like, what the hell is dark matter and dark energy?
start_time: 2591.7200
  end_time: 2595.9600
  distance: 0.7631
-- result 4 --
     title: Alex Filippenko: Supernovae, Dark Energy, Aliens & the Expanding
Universe | Lex Fridman Podcast #137
        id: 51:36
   content:  Do we have any understanding of what the heck that thing is?
start_time: 216.0000
  end_time: 219.0000
  distance: 0.7922
-- result 5 --
     title: Leonard Susskind: Quantum Mechanics, String Theory and Black Holes |
Lex Fridman Podcast #41
        id: 122:831
   content:  That is a big question in physics right now.
start_time: 2374.9000
  end_time: 2377.6200
  distance: 0.8023

================================================================================
Q5a) What are the five most similar podcast episodes to segment '267:476'?

Query:
with episode as (
  select podcast_id, avg(embedding) as embedding
    from segment
    group by podcast_id
)
select p.title, e.embedding <-> q.embedding as distance
  from episode e
  join podcast p on e.podcast_id = p.id
  join segment q on q.id = '267:476'
  where e.podcast_id <> q.podcast_id
  order by distance
```

```
  limit 5;

-- result 1 --
   title: Sara Walker: The Origin of Life on Earth and Alien Worlds | Lex Fridman
Podcast #198
distance: 0.7829
-- result 2 --
   title: Martin Rees: Black Holes, Alien Life, Dark Matter, and the Big Bang | Lex
Fridman Podcast #305
distance: 0.7879
-- result 3 --
   title: Max Tegmark: Life 3.0 | Lex Fridman Podcast #1
distance: 0.7887
-- result 4 --
   title: Sean Carroll: The Nature of the Universe, Life, and Intelligence | Lex
Fridman Podcast #26
distance: 0.7891
-- result 5 --
   title: Nick Bostrom: Simulation and Superintelligence | Lex Fridman Podcast #83
distance: 0.7911

================================================================================
Q5b) What are the five most similar podcast episodes to segment '48:511'?

Query:
with episode as (
  select podcast_id, avg(embedding) as embedding
    from segment
    group by podcast_id
)
select p.title, e.embedding <-> q.embedding as distance
  from episode e
  join podcast p on e.podcast_id = p.id
  join segment q on q.id = '48:511'
  where e.podcast_id <> q.podcast_id
  order by distance
  limit 5;

-- result 1 --
   title: Christof Koch: Consciousness | Lex Fridman Podcast #2
distance: 0.7538
-- result 2 --
   title: Dileep George: Brain-Inspired AI | Lex Fridman Podcast #115
distance: 0.7605
-- result 3 --
   title: Tomaso Poggio: Brains, Minds, and Machines | Lex Fridman Podcast #13
distance: 0.7616
-- result 4 --
   title: Elon Musk: Neuralink, AI, Autopilot, and the Pale Blue Dot | Lex Fridman
Podcast #49
distance: 0.7762
-- result 5 --
   title: Philip Goff: Consciousness, Panpsychism, and the Philosophy of Mind | Lex
Fridman Podcast #261
distance: 0.7872

================================================================================
Q5c) What are the five most similar podcast episodes to segment '51:56'?
```

```
Query:
with episode as (
  select podcast_id, avg(embedding) as embedding
    from segment
    group by podcast_id
)
select p.title, e.embedding <-> q.embedding as distance
  from episode e
  join podcast p on e.podcast_id = p.id
  join segment q on q.id = '51:56'
  where e.podcast_id <> q.podcast_id
  order by distance
  limit 5;

-- result 1 --
   title: Sean Carroll: Quantum Mechanics and the Many-Worlds Interpretation | Lex
Fridman Podcast #47
distance: 0.7767
-- result 2 --
   title: Stephen Wolfram: Fundamental Theory of Physics, Life, and the Universe |
Lex Fridman Podcast #124
distance: 0.8081
-- result 3 --
   title: Donald Hoffman: Reality is an Illusion - How Evolution Hid the Truth |
Lex Fridman Podcast #293
distance: 0.8166
-- result 4 --
   title: Cumrun Vafa: String Theory | Lex Fridman Podcast #204
distance: 0.8173
-- result 5 --
   title: Avi Loeb: Aliens, Black Holes, and the Mystery of the Oumuamua | Lex
Fridman Podcast #154
distance: 0.8255

================================================================================
Q6) What are the five most similar podcast episodes to episode 'VeH7qKZr0WI'?

Query:
with episode as (
  select podcast_id, avg(embedding) as embedding
    from segment
    group by podcast_id
)
select p.title, e.embedding <-> q.embedding as distance
  from episode e
  join podcast p on e.podcast_id = p.id
  join episode q on q.podcast_id = 'VeH7qKZr0WI'
  where e.podcast_id <> q.podcast_id
  order by distance
  limit 5;

-- result 1 --
   title: Tyler Cowen: Economic Growth & the Fight Against Conformity & Mediocrity
| Lex Fridman Podcast #174
distance: 0.1195
-- result 2 --
   title: Eric Weinstein: Difficult Conversations, Freedom of Speech, and Physics |
Lex Fridman Podcast #163
distance: 0.1257
```

-- result 3 --
   title: Michael Malice and Yaron Brook: Ayn Rand, Human Nature, and Anarchy | Lex Fridman Podcast #178
distance: 0.1284
-- result 4 --
   title: Steve Keen: Marxism, Capitalism, and Economics | Lex Fridman Podcast #303
distance: 0.1292
-- result 5 --
   title: Michael Malice: The White Pill, Freedom, Hope, and Happiness Amidst Chaos | Lex Fridman Podcast #150
distance: 0.1304